# Is there a cheaper corner than (degree 5, 4 colors)?

21 of 21 (degree, n_colors) cells x 5 sizes (60-960), 105 removal rows and 63 random rows

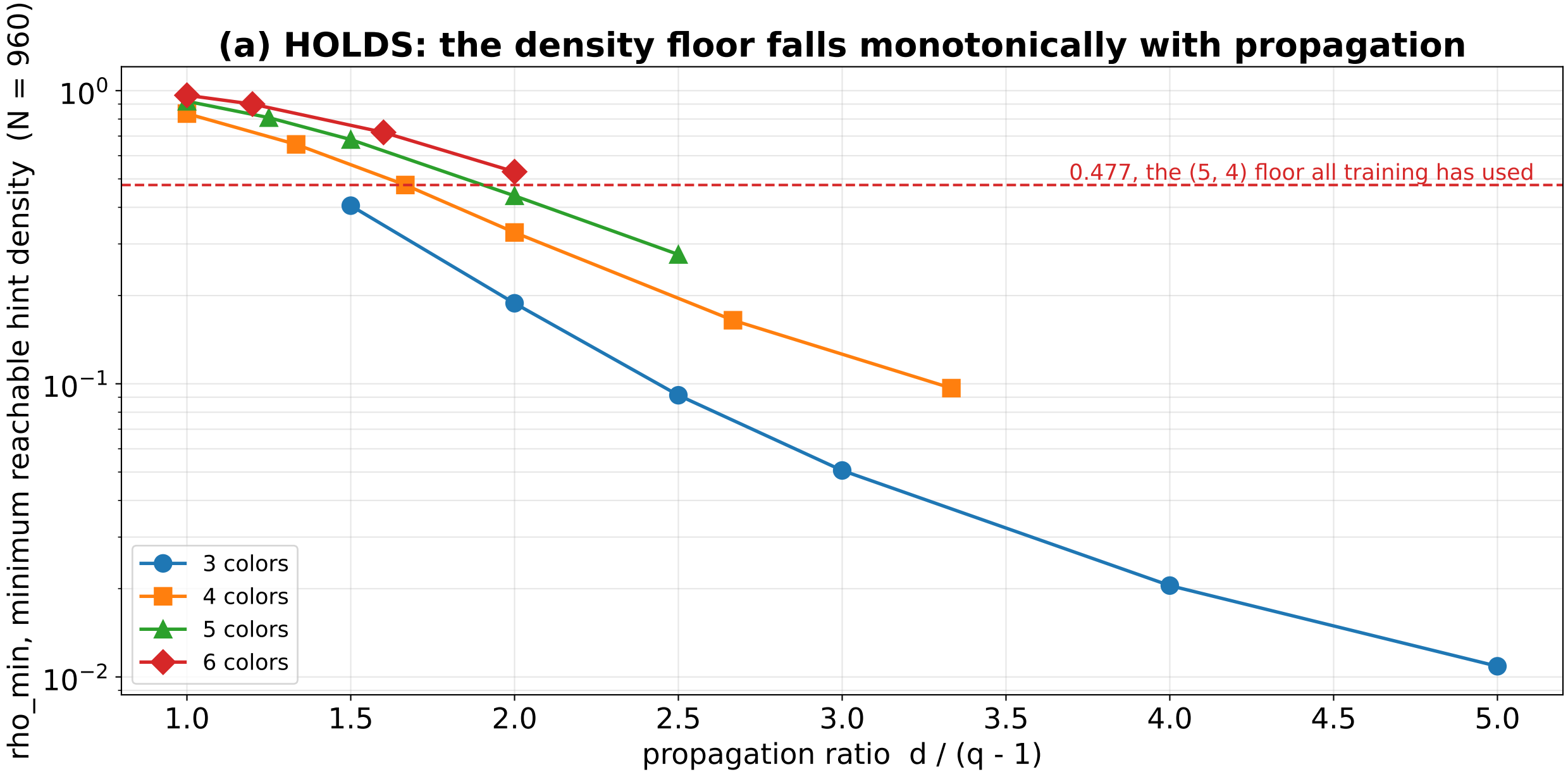
sudoku_dd_control_param_2026-10-25

YES, and at N = 60 it is worth about a factor of 6 in N.

(d=4, q=3) reaches mean deduction depth 14.5 near its density floor at N = 60, against 8.0 for (d=5, q=4) at the same size. The baseline does not reach 14.5 until N ~ 333.

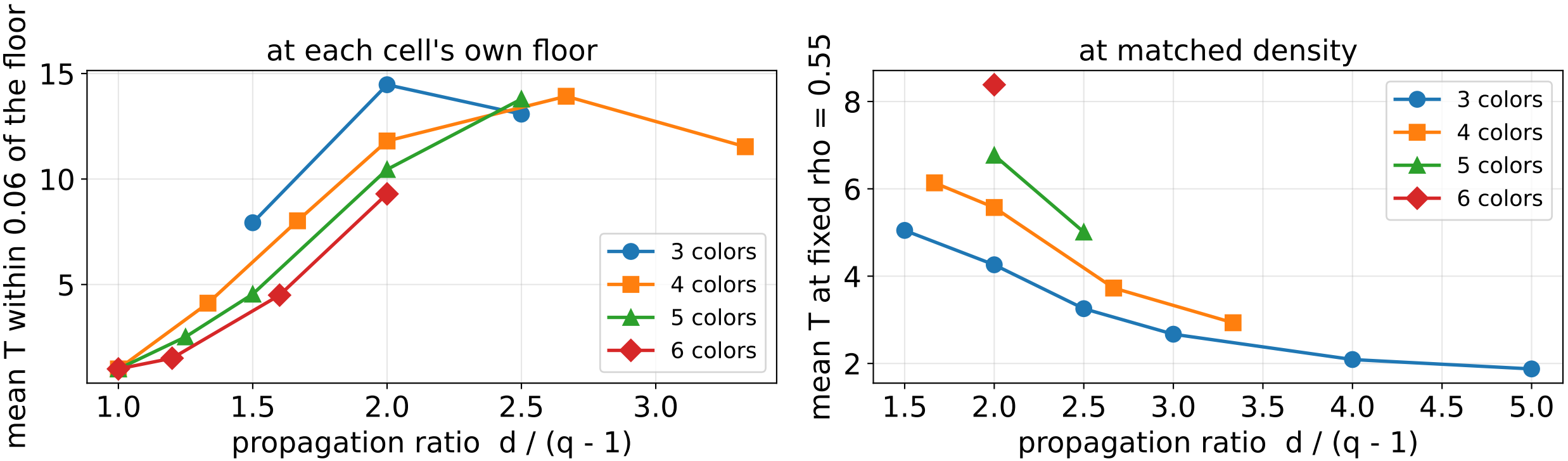
But NOT for the reason predicted, and rho_min is still not a critical point: T grows as a power of N at every density DD can act on.

The propagation ratio d/(q-1) governs how many neighbours there are per colour that has to be excluded before DD can force a node. Prediction (a) was that rho_min falls as that ratio rises; prediction (b) was that the DEEPEST instances live at low ratio. The first holds across three decades; the second is backwards.



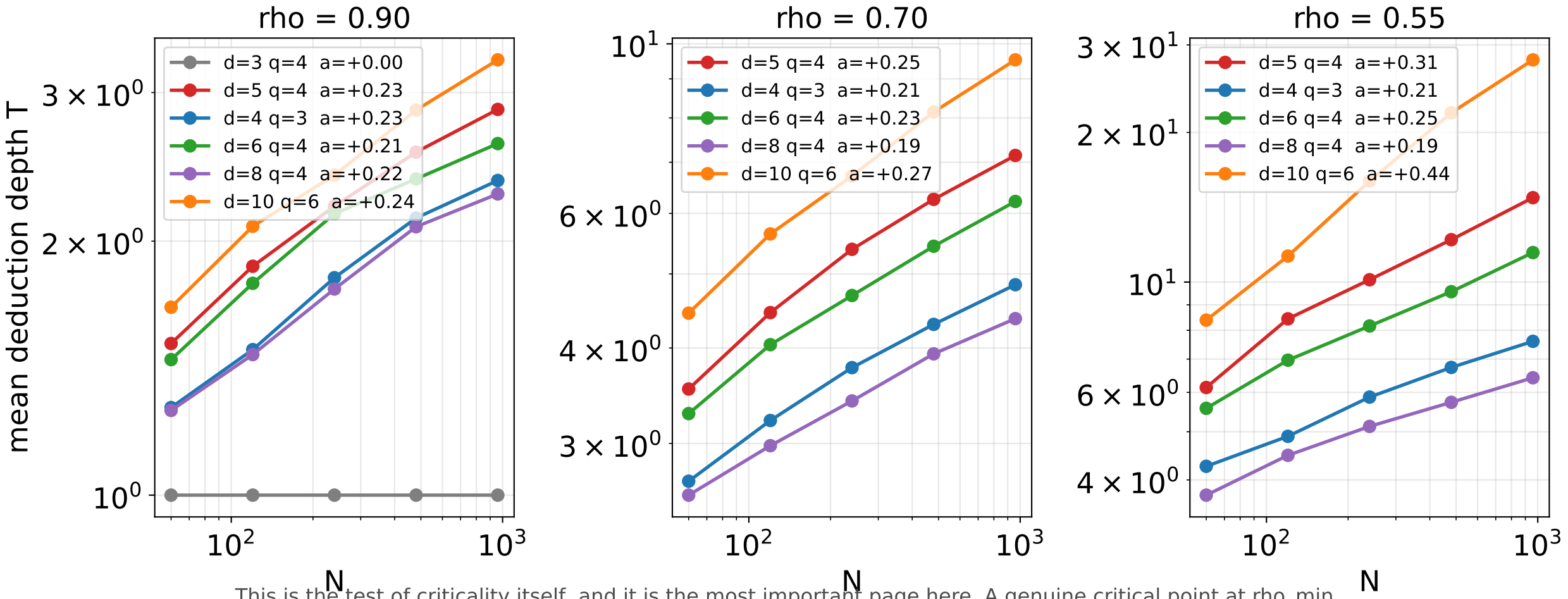

Measured at the largest size swept, N = 960, where rho_min is already asymptotic. It spans 0.84 at ratio 1.00 down to 0.011 at ratio 5.00, a factor of 77, and falls monotonically throughout -- so prediction (c)'s falsifier, that rho_min is flat in (d,q) and there is no cheap corner, is itself refuted. The shallow end is the degenerate case d = q-1, where DD can force almost nothing and greedy removal stalls at 84% hinted.

# (b) FAILS, backwards: depth peaks at HIGH propagation, not low



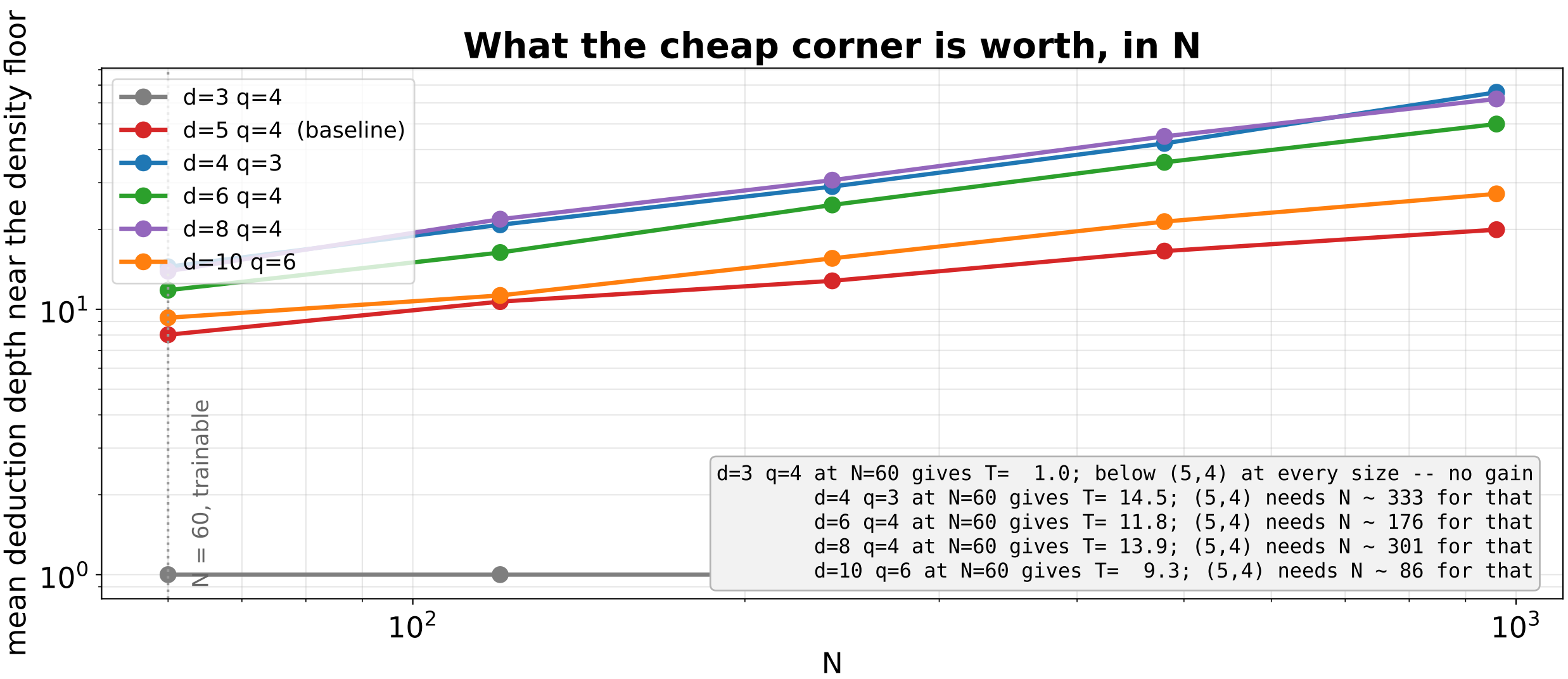

(b) expected a ratio <= 1.2 cell to reach mean T >= 22 at N=96; the rerun moved to multiples of 60 for constructibility, so it is scored at N=60. The verdict is the same at every size: the low-ratio end is d = q-1, the SHALLOWEST cells on the grid, where DD forces nothing and removal stalls at once (T ~ 1). The two panels then invert the role of the COLOR COUNT, and the inversion is the real content of this page. At each cell's own floor (left) depth has an interior optimum in the ratio, peaking at 2.0-2.7 in every q, and MORE colors is SHALLOWER: q=6 tops out at 9.3 against q=3's 14.5. At matched rho=0.55 (right) depth instead falls monotonically with the ratio inside each q, and more colors is DEEPER: 5.0, 6.1, 6.8, 8.4 for q = 3,4,5,6. Both orderings hold unchanged at N=960. The mechanism is the floor. A high-q cell has a high rho_min (0.507 at d=10 q=6 vs 0.204 at d=4 q=3), so a fixed rho=0.55 sits just above its floor while leaving the low-q cell heavily over-hinted. This also corrects the caveat carried from the 12-cell partial grid, that (5,4) was the deepest cell at matched density and the corner bought depth only by moving the floor down: (5,4) ranks 3rd of the 13 cells scorable at rho=0.55, and d=10 q=6 beats it 8.4 to 6.1 at N=60, widening to 28.0 vs 14.8 at N=960, from a floor of 0.507 against the baseline's 0.468. At equal hint density that cell is not a cheaper corner but a deeper one at the same price.

# (d) FAILS: T grows as a power of N at every density DD can act on

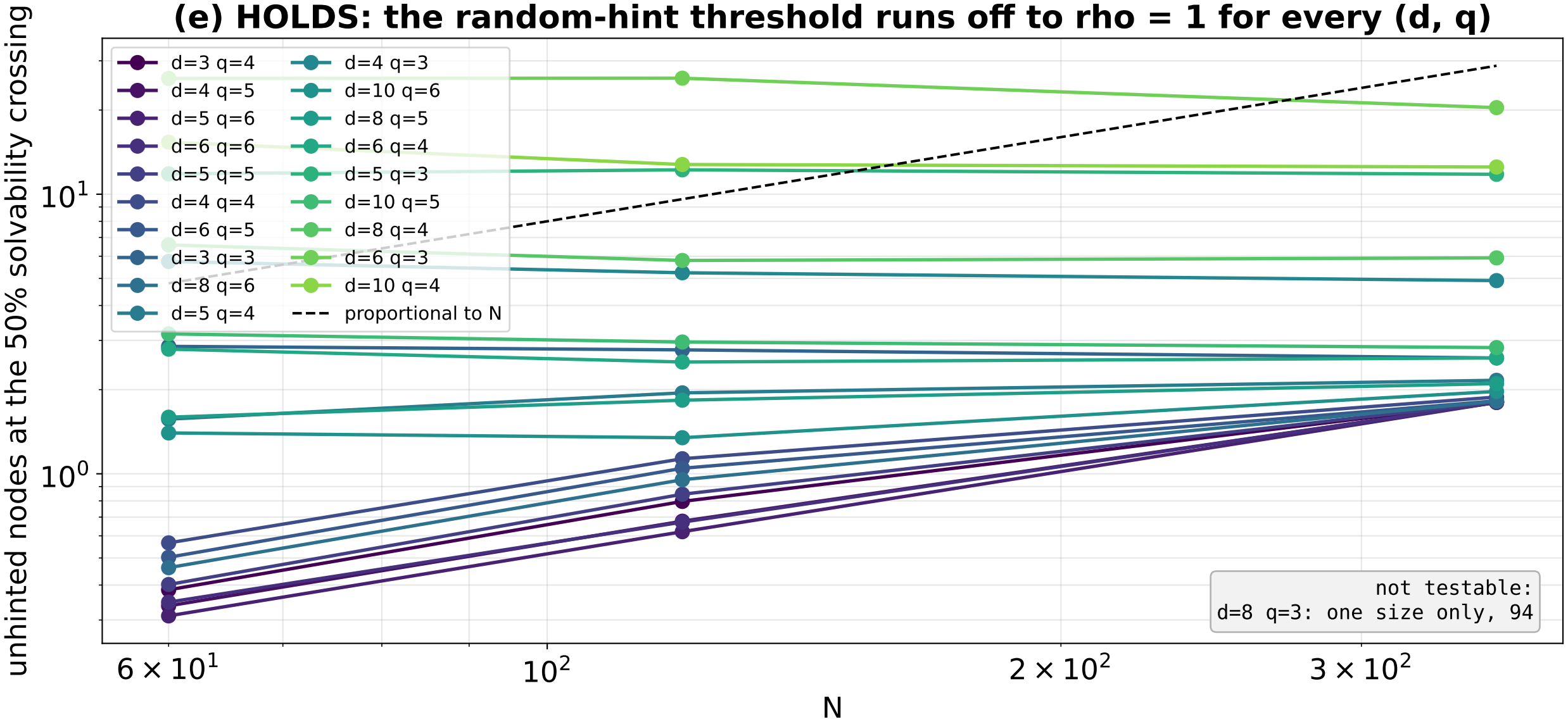



This is the test of criticality itself, and it is the most important page here. A genuine critical point at rho_min requires T to grow with N near the floor while staying N-INDEPENDENT far from it. Instead the exponent a in T ~ N^a is positive at rho = 0.90 in every cell that deduces anything -- it never reaches zero, it only gets smaller. The one exception, d=3 q=4 at a = +0.00, is not evidence for criticality but the opposite: it is pinned at exactly T = 1.0 at every N because d = q-1 lets DD force nothing beyond the first round, so there is no chain to lengthen. Excluding it, rho_min is where greedy removal stops rather than a critical point, across the whole (degree, n_colors) grid and not at (5,4) alone -- a real strengthening of finding 1 rather than a restatement of it.

# What the cheap corner is worth, in N




Read with the caveat from the depth-vs-ratio page: most of these cells reach their depth at their OWN floor, which for (8,4) is rho ~ 0.17 against 0.48 for the baseline, so the instances are far sparser. Whether that is a fair substitute for large N is a question about what the transformer is being asked to learn, not one this sweep answers. Token cost favours those cells twice over: q=3 needs a smaller vocabulary and d=4 a shorter graph section than the baseline's d=5. d=10 q=6 is the exception on both counts, and the reason it was added to this page. Its floor is 0.507 against the baseline's 0.468, so its gain is measured at essentially the SAME density -- the caveat above does not apply to it, and the matched-density panel of the previous page scores it the deepest cell on the grid. The gain is the smallest shown here (N ~ 86 against the baseline's 60, a factor of 1.4) and its token cost runs the wrong way: 6 colors is a larger vocabulary and d=10 twice the baseline's graph section. It buys caveat-free depth, not cheap depth.




If the random-hint ensemble had a finite-density solvability transition, the number of unhinted nodes at the 50% crossing would grow proportionally to N, like the dashed line. It is flat in N in every cell with two or more measurable sizes. The CONSTANT rises steeply with degree -- about 3 at d=3 and 85 at d=8, q=3 -- which is the propagation ratio again, not a transition; the earlier reading of that 85 as pressure on this prediction confused a large constant for N-growth. The cells with one size are nan at the others because the crossing fell below the rho = 0.50 grid floor, not because it is absent: extending the grid downward for ratio >= 4 is the fix, and it is cheap.

# Scorecard against the pre-registered predictions

| | | |
|---|---|---|
| **(a)** | **HOLDS** | rho_min falls monotonically with d/(q-1), 0.84 -> 0.011 over ratio 1.00 -> 5.00. The specific claim rho_min(3,4) > 0.477 was met at 0.836; the d=10 q=4 half awaits the rest of the grid. |
| **(b)** | **FAILS** | Backwards. The ratio <= 1.2 cells are d = q-1, the SHALLOWEST on the grid (T ~ 1, floor at 84% hinted), not the deepest. Depth has an interior optimum in the ratio. |
| **(c)** | **REFUTED** | Its falsifier required rho_min and max T flat within +-20%. They span two decades and a factor of 100. There IS a cheap corner -- which is the useful half of this sweep. |
| **(d)** | **FAILS** | The exponent in T ~ N^a is positive at rho = 0.90 in every cell that deduces at all (+0.19 to +0.25), so nothing is critical at rho_min anywhere on the grid. The sole flat cell, d=3 q=4, is pinned at T = 1.0 for lack of any chain to grow. Generalises finding 1 from (5,4) to all of (degree, n_colors). |
| **(e)** | **HOLDS** | unhinted-at-crossing is flat in N in every cell with two or more sizes. The large constants at high degree are the ratio, not a transition. |

21 of 21 cells and 105 of 126 removal rows are in. (a), (c) and (d) are decided by what is here and the rest of the grid cannot overturn them. (b) could still be rescued by a ratio <= 1.2 cell at higher q, which is what remains -- treat it as very likely failed rather than settled.